

Cells: Structure, Transport & Division

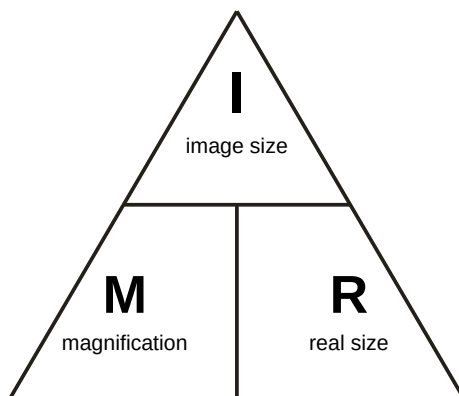
Quick reference sheet · Biology & Maths · keep handy for revision

1. Microscopy & magnification

An electron microscope has far higher resolving power than a light microscope (up to $\times 1,000,000$ vs $\times 1,000$), so it reveals much finer sub-cellular detail.

magnification = size of image $\div$ size of real object or **$M = I / R$**

Image and real size must be in the **same units** before dividing. 1 mm = 1000 μm .



Cover **I** $\rightarrow$ M next to R $\rightarrow I = M \times R$. Cover **M** $\rightarrow$ I over R $\rightarrow M = I / R$. Cover **R** $\rightarrow$ I over M $\rightarrow R = I / M$.

2. Cell structures

Animal & plant cells are **eukaryotic** (genetic material in a nucleus). Bacterial cells are **prokaryotic** — smaller, with a single DNA loop and plasmids, not in a nucleus.

Structure	Animal	Plant	Bacterial	Function
Nucleus	✓	✓	—	Contains genetic material
Cytoplasm	✓	✓	✓	Most chemical reactions
Cell membrane	✓	✓	✓	Controls entry/exit
Mitochondria	✓	✓	—	Respiration (releases energy)
Ribosomes	✓	✓	✓	Protein synthesis
Cell wall	—	✓ (cellulose)	✓	Strengthens the cell
Chloroplasts	—	✓	—	Photosynthesis
Permanent vacuole	—	✓	—	Cell sap, keeps cell rigid
DNA loop & plasmids	—	—	✓	Genetic material (no nucleus)

3. Transport across membranes

Process	What moves	Direction	Energy	Example
Diffusion	Particles	High → low conc.	No	O ₂ alveoli → blood
Osmosis	Water only	High → low water conc., partially permeable membrane	No	Water into root hairs
Active transport	Dissolved substances	Low → high conc. (against gradient)	Yes	Mineral ions into root hairs

4. Cell division: mitosis & meiosis

Human body cells have **46 chromosomes** (23 pairs). **Mitosis** makes 2 identical cells for growth & repair. **Meiosis** makes 4 genetically different gametes, each with half the chromosome number (23). Fertilisation restores the full number.

	Mitosis	Meiosis
Purpose	Growth & repair	Making gametes
Divisions	One	Two
Cells produced	2	4
Chromosome no.	Same as parent (46)	Half the parent (23)
Genetically identical?	Yes	No — all different

5. Cell differentiation & stem cells

Early embryo cells are unspecialised **stem cells** that can become any cell type. As development continues, most cells **differentiate** into specialised cells (e.g. nerve, muscle, red blood cell) and cannot change type again. Most adult tissues still keep some stem cells to replace damaged cells.